Lect.9 Machine Learning Classification

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Lect. 9 Machine Learning Classification

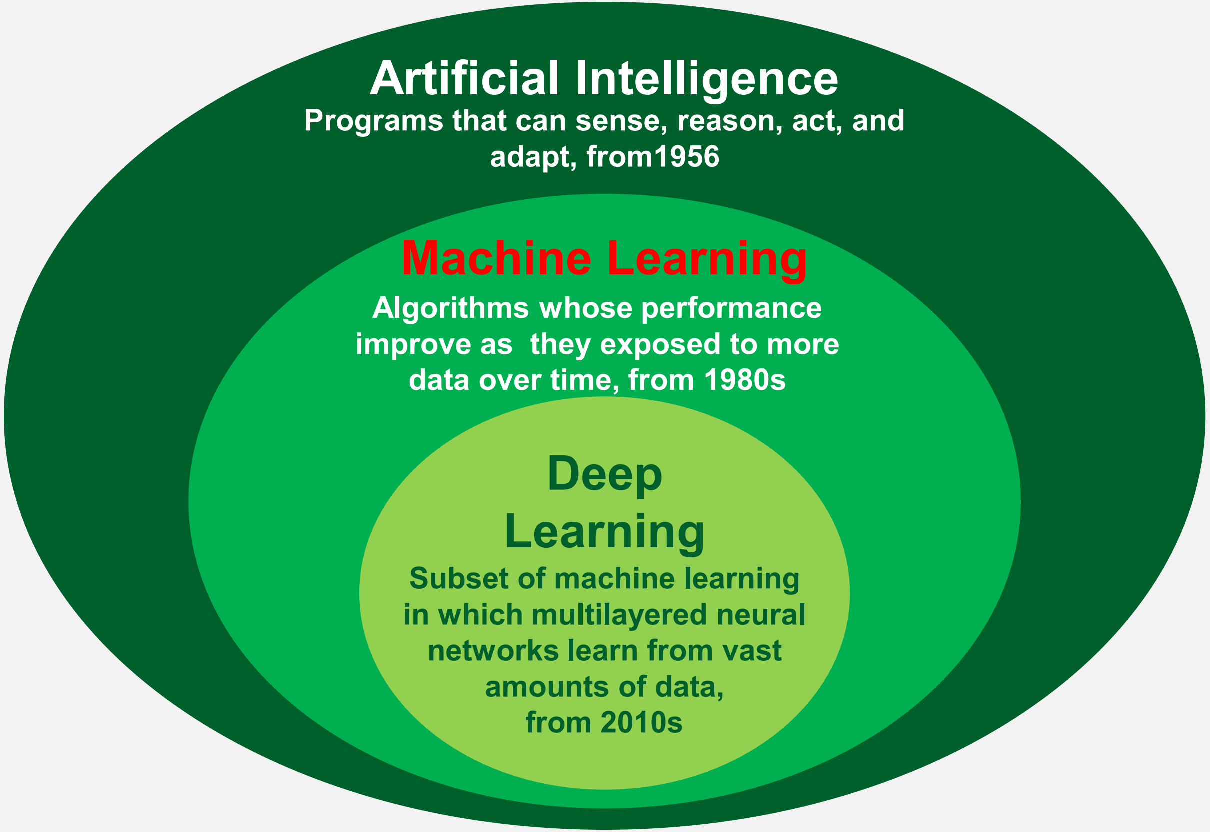
- □ What is Machine Learning
- □ Random Forest Classification
  - ➢ Decision Tree
  - ➢ Ensemble methods
  - ➢ Random Forest
- □ Cases

What is Machine Learning

Machine learning is a branch of artificial intelligence and computer science which focuses on the use of data and algorithms to imitate the way that humans learn, gradually improving its accuracy.

- An important component of the growing field of big data science.
- Through the use of statistical methods, algorithms are trained to make classifications or predictions, uncovering key insights within data mining projects.

Machine Learning vs. Artificial Intelligence vs. Deep learning



- Machine learning, deep learning, and neural networks are all sub-fields of artificial intelligence.
- Deep learning is actually a sub-field of machine learning, and deep neural networks is a sub-field of deep learning.

How Machine Learning works

Machine learning algorithm into three main parts：

- **A Decision Process**: In general, machine learning algorithms are used to make a prediction or classification. Based on some input data, which can be labelled or unlabeled, your algorithm will produce an estimate about a pattern in the data.
- **An Error Function**: An error function serves to evaluate the prediction of the model. If there are known examples, an error function can make a comparison to assess the accuracy of the model.
- **An Model Optimization Process**: If the model can fit better to the data points in the training set, then weights are adjusted to reduce the discrepancy between the known example and the model estimate. The algorithm will repeat this evaluate and optimize process, updating weights autonomously until a threshold of accuracy has been met.

Three Primary Categories of Machine Learning Methods

Supervised, unsupervised and semi-supervised machine learning:

- **Supervised machine learning**
  - Supervised machine learning (or supervised learning), is defined by its use of labeled datasets to train algorithms that to classify data or predict outcomes accurately.
  - As input data is fed into the model, it adjusts its weights until the model has been fitted appropriately.
  - This occurs as part of the cross validation process to ensure that the model avoids overfitting or underfitting.
  - Some methods used in supervised learning include neural networks, naive Bayes, **<u>linear regression</u>**, logistic regression, **<u>random forest</u>**, support vector machine (SVM), and more.

Three Primary Categories of Machine Learning Methods

Supervised, unsupervised and semi-supervised machine learning:

- **Unsupervised machine learning**
  - It is also known as unsupervised learning, uses machine learning algorithms to analyze and cluster unlabeled datasets.
  - This category of algorithms discover hidden patterns or data groupings without the need for human intervention. Its ability to discover similarities and differences in information make it the ideal solution for exploratory data analysis, cross-selling strategies, customer segmentation, image and pattern recognition.
  - It's also used to reduce the number of features in a model through the process of dimensionality reduction; principal component analysis (PCA) and singular value decomposition (SVD,奇异值分解) are two common approaches for this.
  - Other algorithms used in unsupervised learning include neural networks, ***k*-means clustering**, **ISODATA,** probabilistic clustering methods, and more.

Three Primary Categories of Machine Learning Methods

Supervised, unsupervised and semi-supervised machine learning:

➢ **Semi-supervised learning**

- Semi-supervised or few-shot learning offers a happy medium between supervised and unsupervised learning.
- During training, it uses a smaller labeled data set to guide classification and feature extraction from a larger, unlabeled data set.
- Semi-supervised learning can solve the problem of having not enough labeled data (or not being able to afford to label enough data) to train a supervised learning algorithm.

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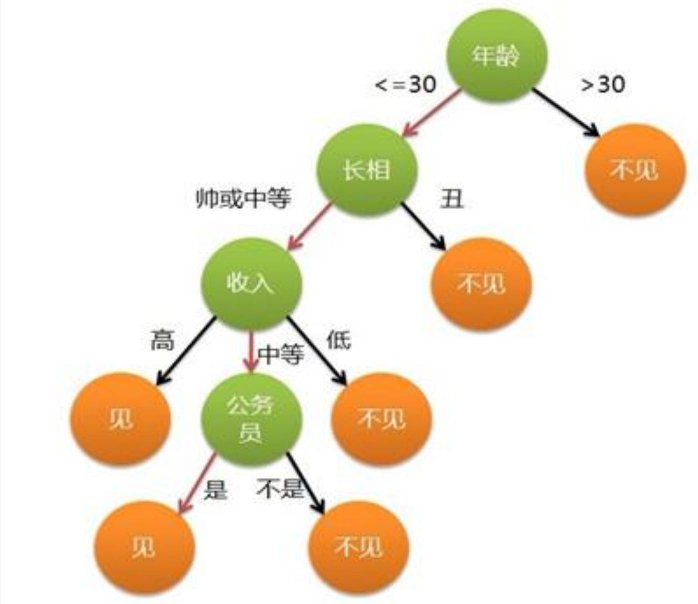
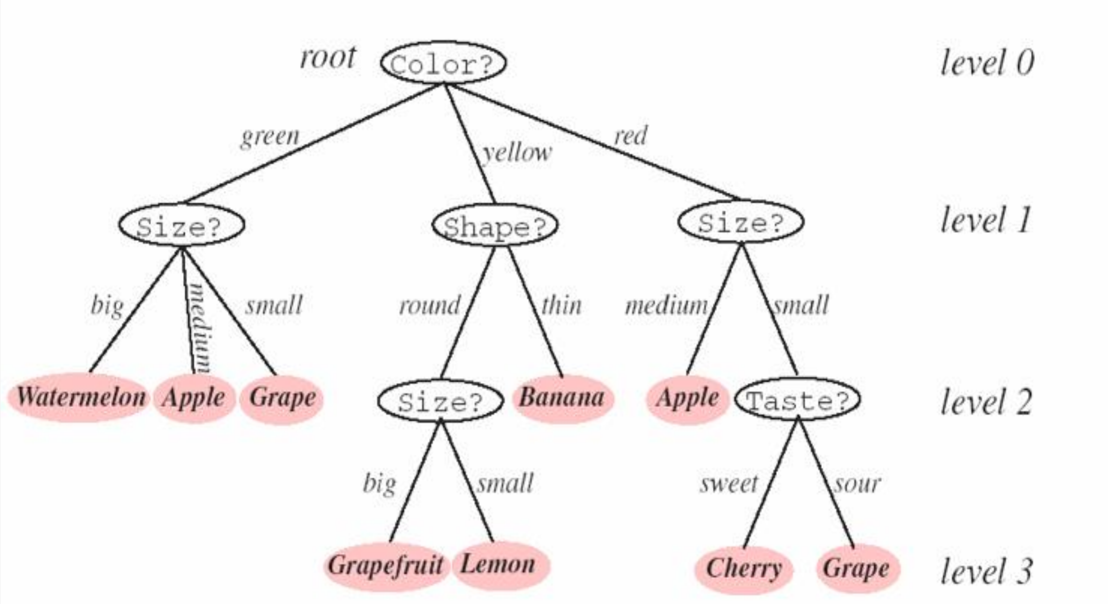
Decision Tree

- A **decision tree** is a decision support tool that uses a tree-like model of decisions and their possible consequences, including chance event outcomes, resource costs, and utility.
- It is one way to display an algorithm that only contains conditional control statements.
- It is commonly used in **image classification**, operations research, operations management, etc.

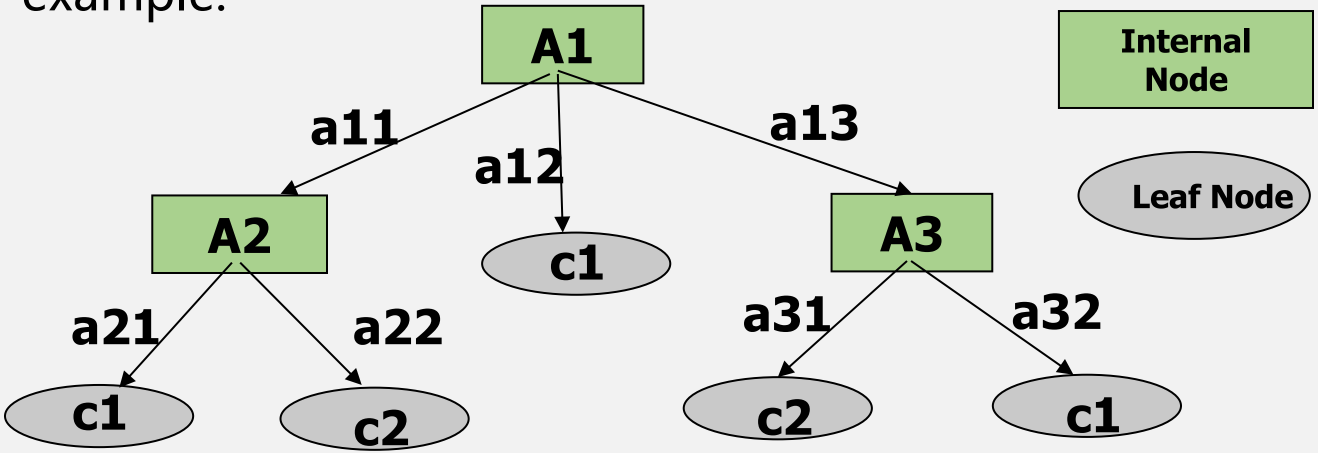




Decision Tree

Two types of nodes in a decision tree:

- Internal node: Splits data into different branches according to the different values the corresponding attribute can take.
- Terminal/Leaf node: Decides the class assigned to the example.



Mechanism to Construct Decision Trees

Different ways to construct decision trees from data,

We concentrate on the top-down, **greedy search approach** **(**贪婪搜索算法):

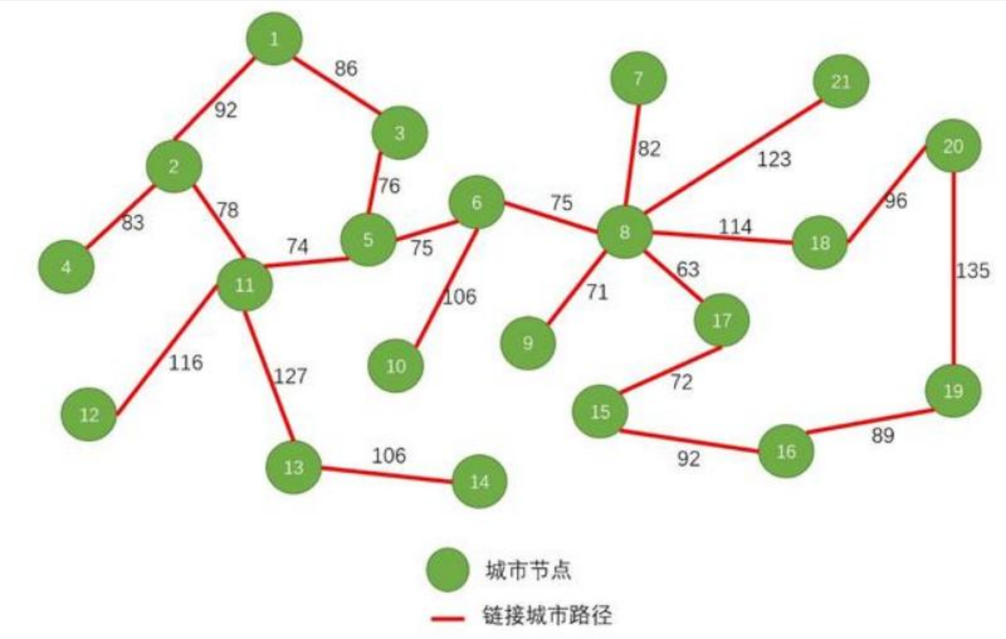
Basic idea:

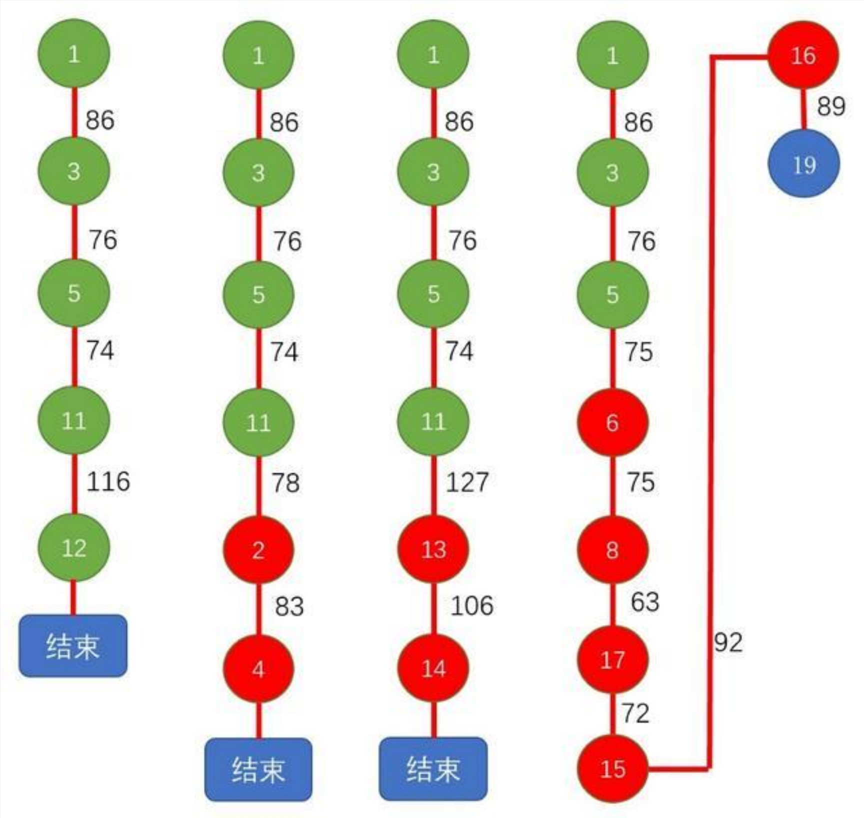
- Choose the best attribute *a** to place at the root of the tree.
- Separate training set D into subsets $\{D_1, D_2, .., D_k\}$ where each subset D_i contains examples having the same value for *a**.
- Recursively apply the algorithm on each new subset until examples have the same class or there are few of them.

Decision Tree

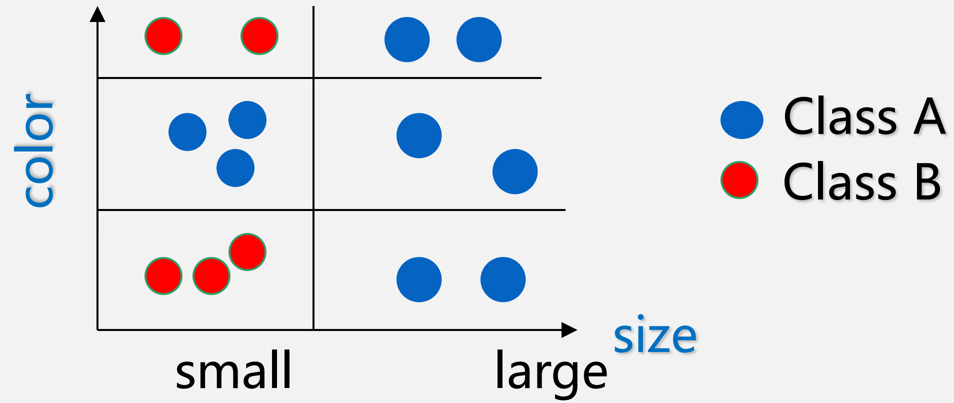
Greedy search approach

Find the shortest path from node 1 to node 19



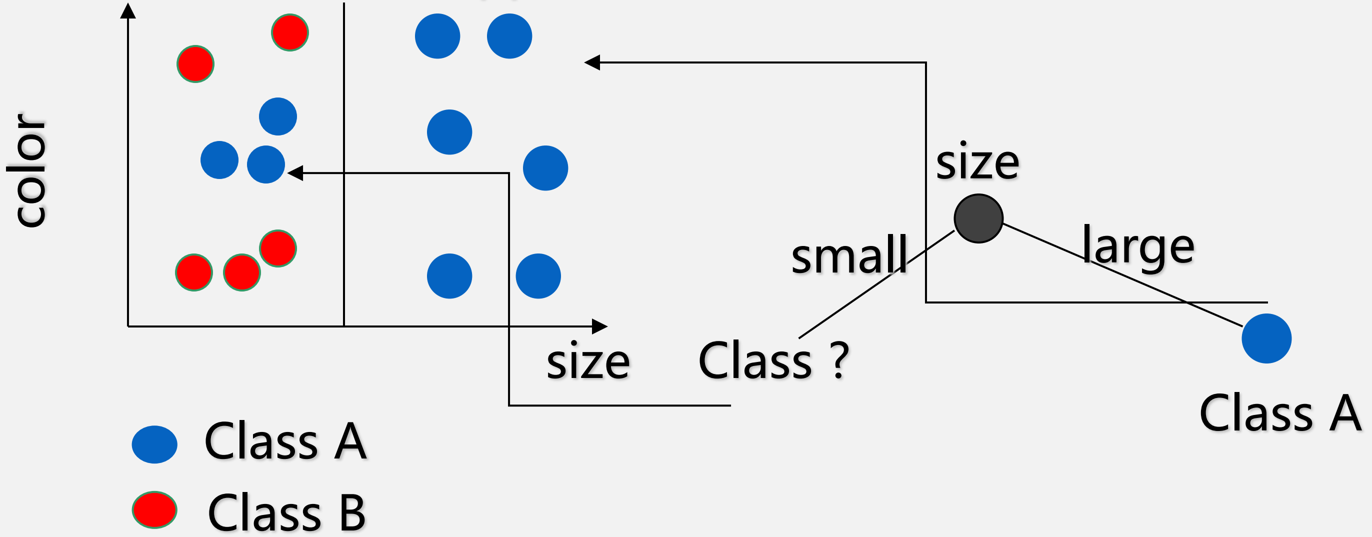


Example 1

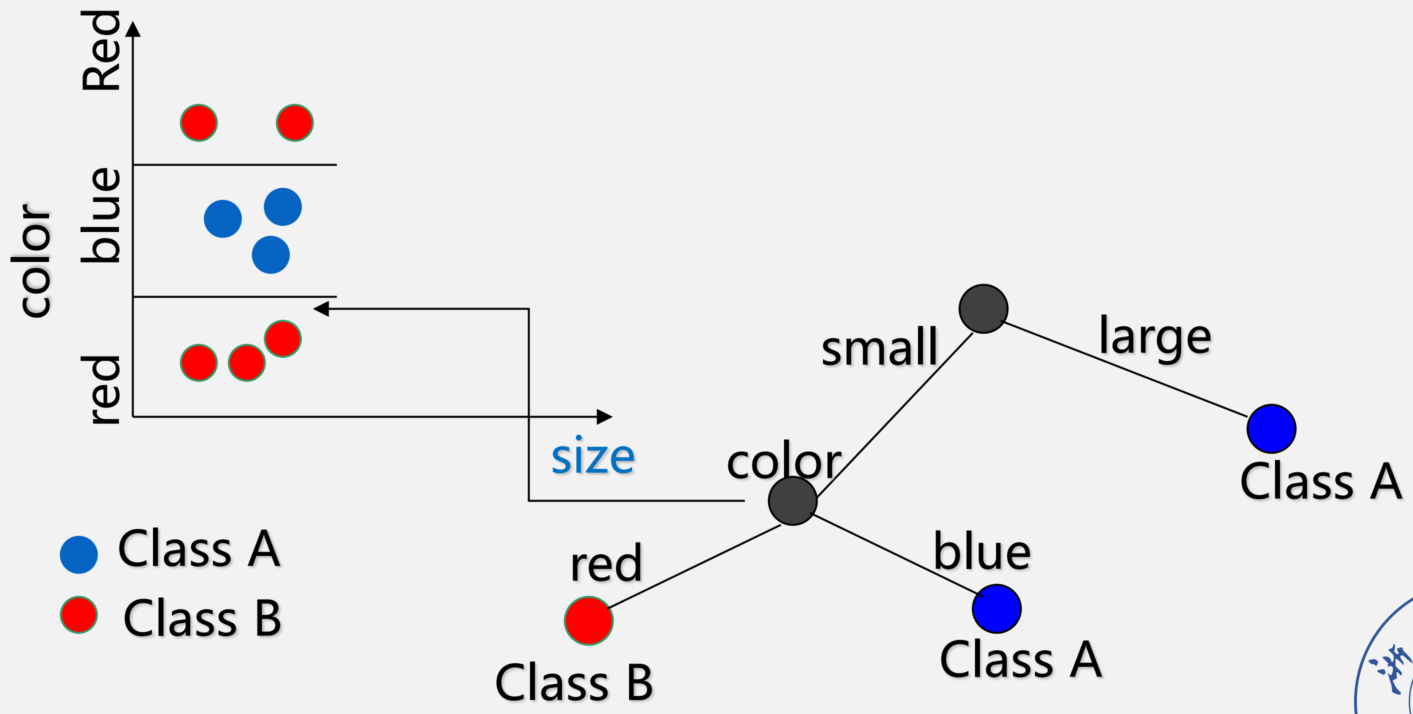


Example 1

Suppose we choose **size** as the best attribute:



Example 1



Red
blue
red
color
size
Class A
Class B
color
small
large
red
blue
Class A
Class B
Class A

Example 2:
A Decision Tree Classifier

Lu et al. A Decision-tree Classifier for Extracting Transparent Plastic-mulched Landcover from Landsat-5 TM Images. IEEE Journal of Selected Topics in Applied Earth Observation and Remote Sensing, 2014.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

A Decision-Tree Classifier for Extracting Transparent Plastic-Mulched Landcover from Landsat-5 TM Images

Lizhen Lu, Liping Di, *Senior Member, IEEE*, and Yanmei Ye

***Abstract*—The area of plastic-mulched landcover (PML), an important agricultural landscape, is increasing rapidly at a rate of 20% per year globally. However, the spatial and temporal distributions of PML have been poorly understood because of lack of effective technology to extract PML for large geographic areas. This paper presents a decision-tree classifier for extracting the transparent PML information from Landsat-5 TM images. The classifier was built with rules obtained from analyzing the spectral characteristics of transparent PML on Landsat-5 TM images, covering the study area of 14 000 km^2 in Xinjiang, the largest PML-based cotton plantation provinces in China. Then, the classifier was applied at the study area for years 1998, 2007, and 2011. Results indicate that the classifier successfully extracted the PML from Landsat-5 TM images at overall accuracies of 97.82%, 85.27%, and 95.00% and Kappa coefficients of 0.9782, 0.80, and 0.93 for years 2011, 2007, and 1998, respectively. The results also imply that the decision-tree classifier is temporally stable and can be applied in different years. Visual comparison of the results with the high-spatial resolution images on Google Earth also shows that detected locations of PML are correct. The study shows that the classifier is an effective method for extracting PML for large geographic areas from Landsat-5 TM. Because of the long history of global coverages as well as free availability of Landsat-5 TM images, it is feasible to map the spatio–temporal dynamics of PML over large geographic areas with the technologies presented in this paper.**

***Index Terms*—Decision-tree classifier, Landsat TM, plastic-mulched landcover (PML), Xinjiang cotton region.**

I. Introduction

SINCE the first use of plastic film in agriculture was in 1948 [1], plastic covering has been used extensively in the cultivation of vegetables (e.g., tomato and beans), fruits (e.g., strawberries), and crops (e.g., cotton, wheat, and rice) [2]. The purpose of using plastics cover in agriculture is to increase the productivity by mitigating the threats of coldness, heat, drought, wind, insects, and crop diseases [3]. Plasticulture, in a broad sense, is defined as the use of plastics in agriculture, including both plant and animal production [3]–[6]. The main types of plastic landcover include greenhouse or walk-in tunnels, small tunnels, and plastic mulch (Fig. 1). There are mainly two types of plastic film used as mulch: 1) the transparent and 2) the black. This study concentrated on plastic mulch with transparent film, the dominate type of plastic landcover.

Plasticultural area has been rapidly expanding worldwide in the past 2 decades and now represents an important cultivated landscape. It was reported that plasticultural area has been expanded at the rate of about 20% per year on average globally over the last decade [2]. In China, this is especially true due to the large-scale shift in land use from cereal crops to more profitable vegetables and cash crops in the last 2 decades [7]. The plasticultural area in China has grown from 4200 ha in 1981, 11 966 873 ha in 2003, to 28 000 000 ha in 2010, with 99% of them transparent plastic mulch [8]–[10].

The large-scale landcover change introduced by plasticulture must have impacts on climate, ecosystem, and environment regionally and globally because it alters the energy balance and water cycles on the land surfaces, deteriorates the soil structure, and reduces the biodiversity. For example, transparent plastic-mulch film allows visible lights to penetrate, but blocks outgoing long-wave radiation and thus causes the greenhouse effect [11], whereas black plastic-mulch film has opposite effects. Plastic-mulch film also prevents the evaporation of water from the soil and reduces energy consumption. The plastic film wastes in the soil harm the plant root system development [12], [13]. Plasticulture also changes pollination of plants and distribution of insects and bird and has a negative effect on biodiversity [14]. Therefore, mapping and monitoring the plasticulture on a large geographic area are important both scientifically and socio-economically.

Remote sensing is the only feasible approach for monitoring plasticulture and understanding its impacts on climate and eco-environment in a large geographic area (e.g., whole China or whole East Asia). In recent years, there have been a limited number of studies on remote sensing of plasticulture by using high (at meter level) spatial resolution images. Carvajal *et al.* [15] proposed an artificial intelligence neural network to detect greenhouse using 2.44 m-resolution QuickBird imagery. With analysis of data collected by ground spectrometer, Levin *et al.* [16] claimed that white and transparent plastic-mulch films have three absorptions centered at 1218, 1732, and 2313 nm that are not affected by dust, rinse, and surface factors. They classified 1 m-resolution AISA-ES hyper-spectral imagery to reach a

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Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

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Example 2

- **What is PML?**

  Plastic covered farmland, or plasticulture, in the broad sense, is defined as the use of plastics in agriculture.

- **What types?**

  Plastic-mulched landcover (PML, 95%), greenhouse or walk-in tunnels, and small tunnels are the three main types of plasticulture.

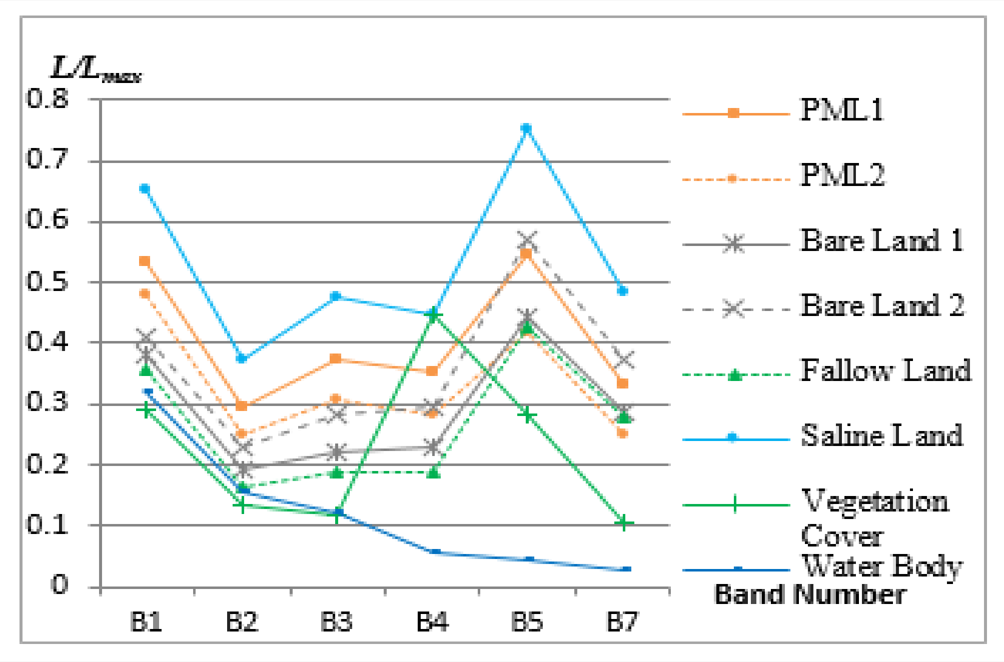
- What purposes?

  The purposes of using plastic cover in agriculture are to mitigate the threats of insects, crop diseases, coldness, heat, drought, and strong rainfall and wind, and to improve the productivity.

- PML, the dominate type of plasticulture in term of area coverage

Plasticulture landscapes in Asia

Example 2: A Decision Tree Classifier



归一化水体指数：

NDWI: Normalized Difference Water Index

NDWI=(Green-NIR)/(Green +NIR)

归一化植被指数 NDVI：

Normalized Difference Vegetation Index

NDVI=(NIR-Red)/(NIR-Red)

覆膜农田指数 PMLI：

Plastic Mulched Landcover Index:

PMLI=(MIR-Red)/(MIR-Red)

Example 2: A Decision Tree Classifier

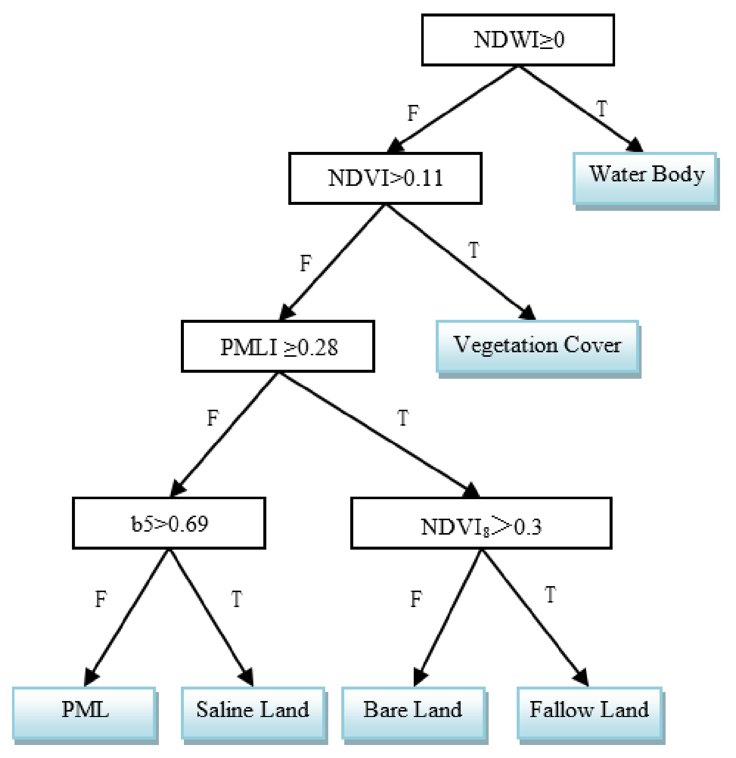
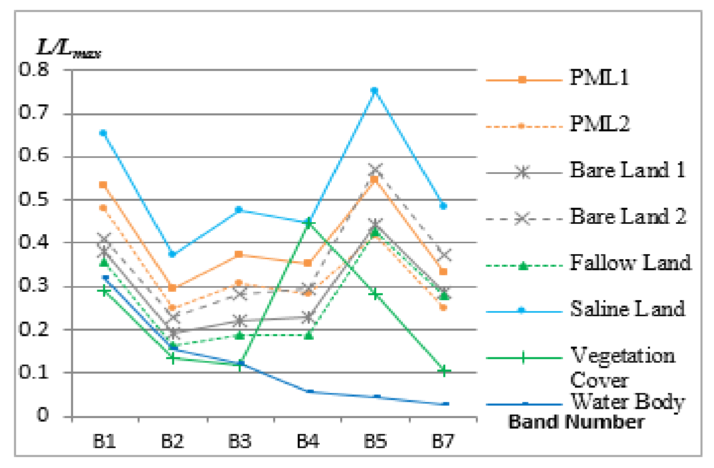


Fig. The decision tree classifier for extracting PML

(Note: T stands for True, F False, and $NDVI_8$ the NDVI value in the peak of a growing season, e.g., August. All other variables in the decision tree are derived from the TM image in the early growing season, e.g., middle to late May in the study area)



$NDVI_8$--Bare land and fallow land have similar spectral signatures in May but can be differentiated by the Landsat images in July/August (peak of the growing season) since fallow land grows a lot of weeds but bare land don't.

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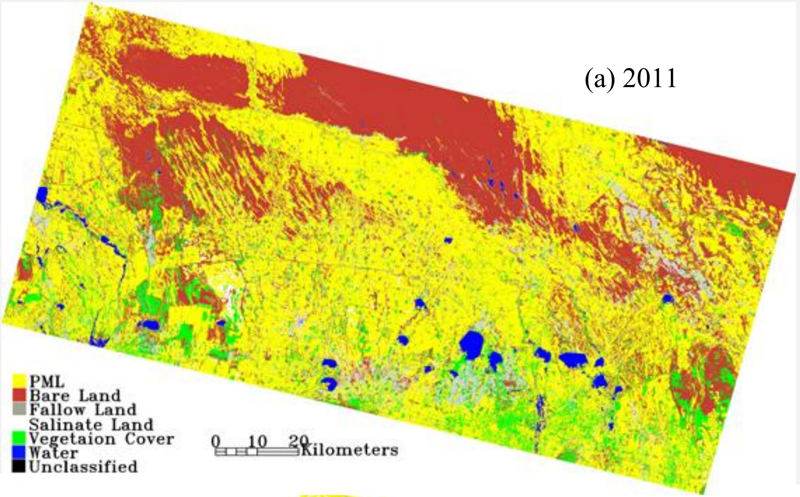
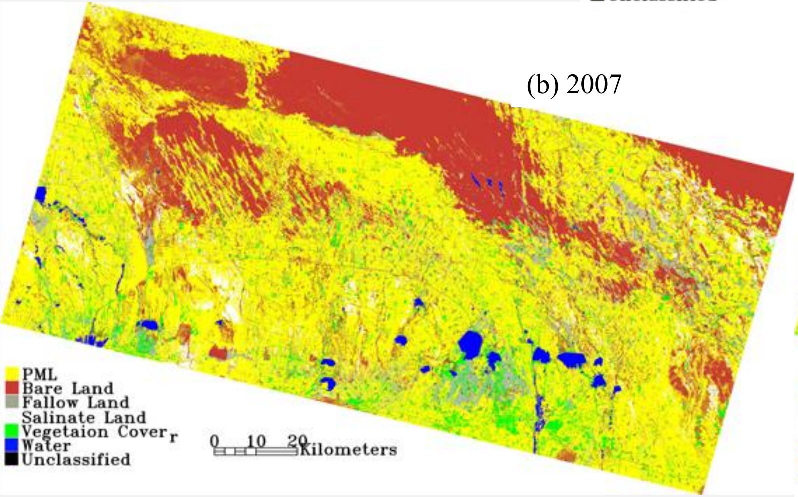
Example 2:

The Classification Results of the Decision Tree





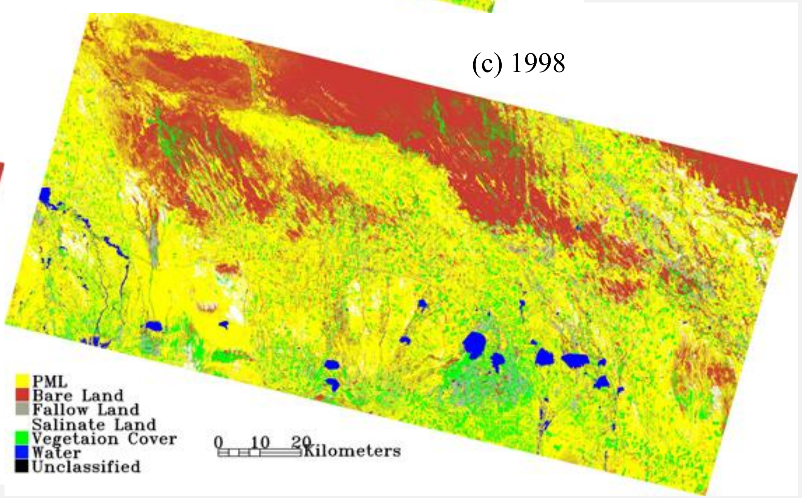


Fig. Classification results from the decision tree classifier

Example 2: The Classification Results of the Decision Tree

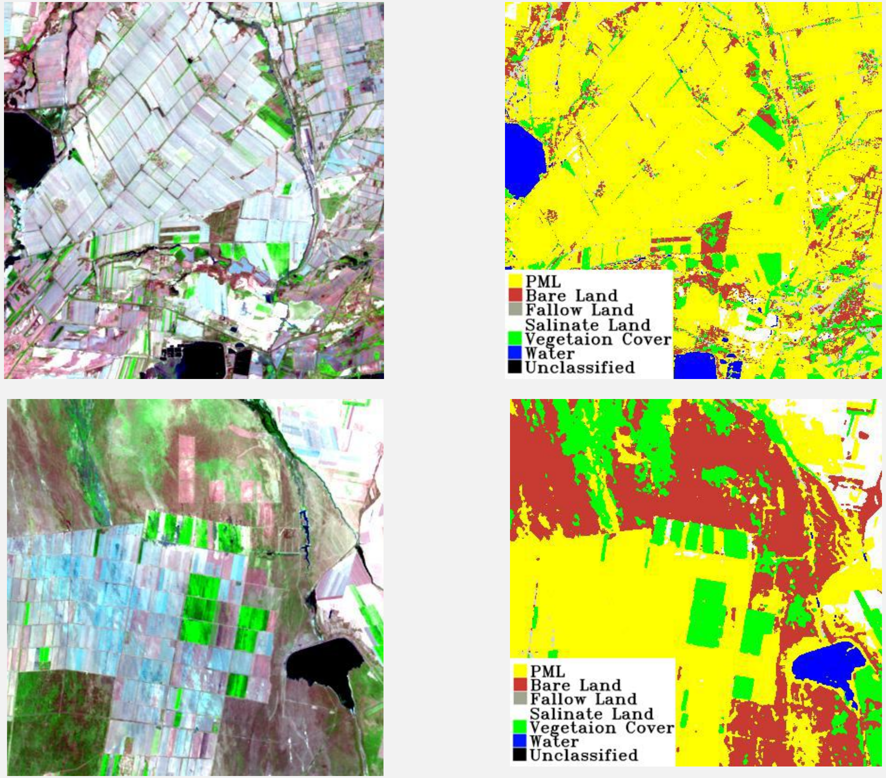


Fig. Comparison of classification results and their corresponding color-composite images in 2011

Decision Tree: Splitting Data

- The above examples show: the decision trees classify the unknown data layer by layer according to the classification attributes/features, until a leaf node is reached.

What attribute/feature is the best to split the data?

- A variety of methods can be used to split data, while each method has its own advantages and disadvantages.
- The complexity of the data set can be measured, and the complexity of classified results by different attributes/features are compared.
  - If the complexity of the classified result by a certain attribute/feature is reduced more, this feature is the best.

How to measure the complexity of data or classification results?

ID3 Using information entropy and information gain (IG) to Split Data

- **ID3**, which is the typical decision tree algorithm, and also the basis for C4.5 and C5.0 algorithms, uses **information entropy (信息熵)** and **information gain (IG，信息增益)** to measure which attribute/feature is the best to split the data.
  - The concept of information entropy was introduced by Claude Shannon in his 1948 paper “A Mathematical Theory of Communication”, and is sometimes called Shannon entropy in his honor.
- The **entropy** of a random variable is the average level of "information", "surprise", or "uncertainty" inherent in the variable's possible outcomes.
- A measure of uncertainty or **entropy** that is associated to a random sample set S is defined as:

$$H(S) = -\sum_{i=1}^{K} p_i \log_2 p_i$$

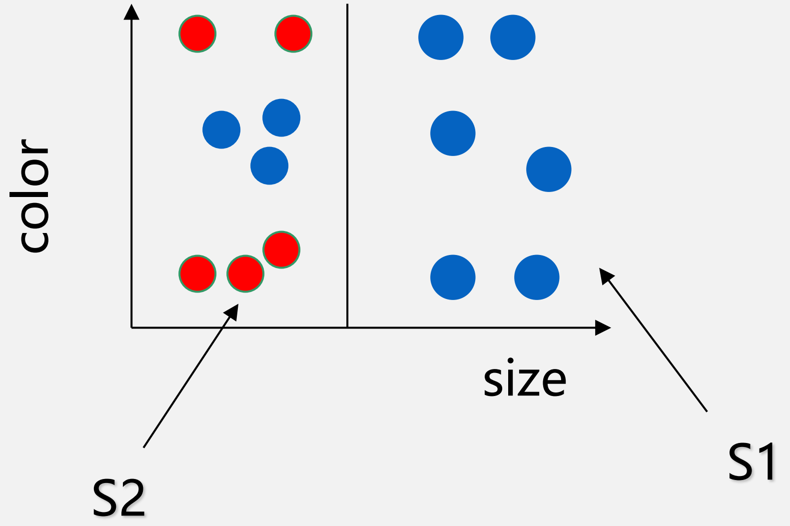
where p_i refers to the **probability** of the i-th possible outcome or event in a random process

The greater the entropy, the more complex the data or information.

Shannon, Claude E. (July 1948). "A Mathematical Theory of Communication". Bell System Technical Journal. ***27*** *(3): 379–423.*

Splitting Data based on Entropy

Let's go back to our previous sample:



Size divides the samples into two groups.

S1 = { 6T, 0F}

S2 = { 3T, 5F}

$$H(S) = -\sum_{i=1}^{K} p_i \log_2 p_i$$

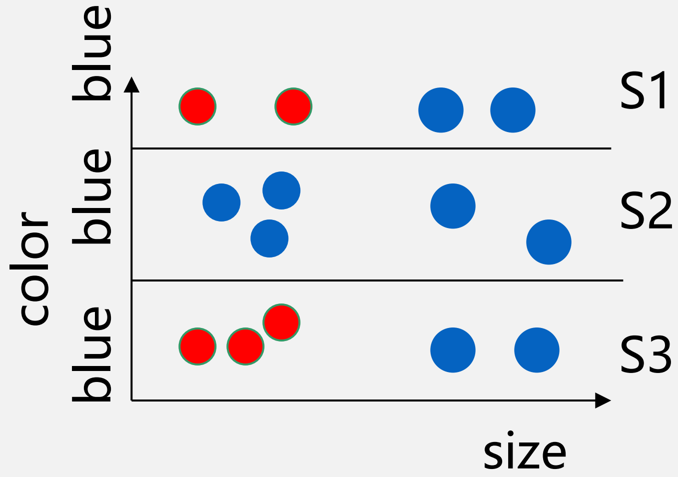
$H(S1) = 0$

$H(S2) = -(3/8)\log_2(3/8)$
$-(5/8)\log_2(5/8)$
$=0.954$

$H(S) = -(9/14)\log_2(9/14)$
$-(5/14)\log_2(5/14)$
$=0.94$

Splitting Data based on Entropy

Let's go back to our previous sample:



Color divides the samples into three groups.

S1 = { 2T, 2F}
S2 = { 5T, 0F}
S3 = { 2T, 3F}

$$
\begin{aligned}
H(S1) &= 1 \\
H(S2) &= 0 \\
H(S3) &= -(2/5)\log_2(2/5) \\
&\quad -(3/5)\log_2(3/5) \\
&= 0.971
\end{aligned}
$$

$$
\begin{aligned}
H(S) &= -(9/14)\log_2(9/14) \\
&\quad -(5/14)\log_2(5/14) \\
&= 0.94
\end{aligned}
$$

Information Gain: IG(X)

Information gain (IG) represents the difference between the two information entropy. It is used to select the feature in the decision tree algorithm.

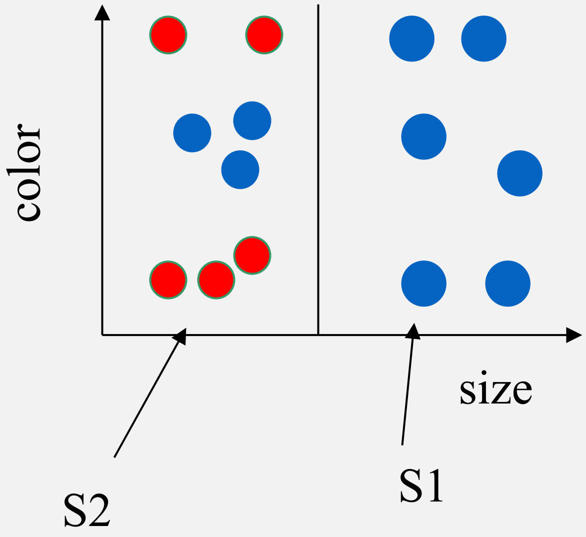
IG of example set S over attribute X:

$$IG(S,X) = H(S) - \sum_{v \epsilon Values(X)} (S_v/S)H(S_v)$$

- *H(S)* : the entropy of all examples *S*
- $Values(X)$: all possible values of attribute *X*
- $H(S_v)$ is the entropy of subsample S_v
- Then the "best" test value is the choice that maximizes IG(S, X)

The greater the information gain (IG), the better the selectivity of the feature is.

Information Gain: IG(S,X)



H(S1) = 0

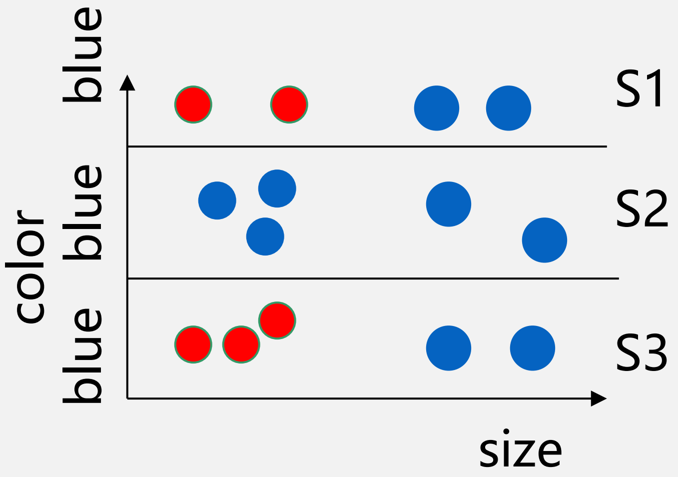
H(S2) = $-(3/8)\log_2(3/8)$
$-(5/8)\log_2(5/8)$
$=0.954$

H(S) = $-(9/14)\log_2(9/14)$
$-(5/14)\log_2(5/14)$
$=0.94$

$$IG(S,X) = H(S) - \sum_v \left(\frac{S_v}{S}\right) H(S_v)$$

=0.94-6/14*0-8/14*0.954
=0.395

Information Gain: IG(S,X)



$H(S1) = 1$

$H(S2) = 0$

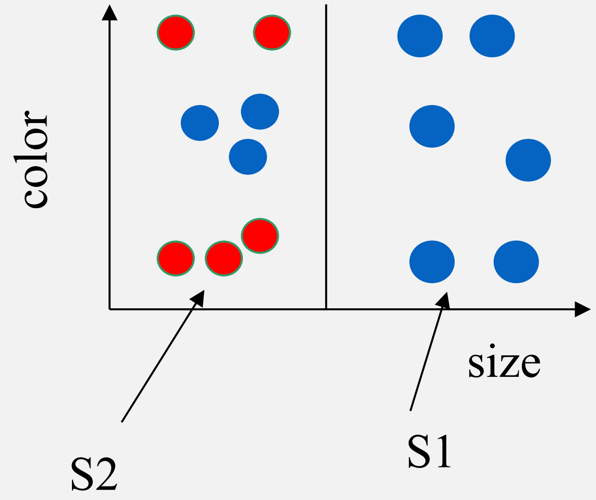
$$H(S3) = -(2/5)\log_2(2/5) \\ -(3/5)\log_2(3/5) \\ =0.971$$

$$H(S) = -(9/14)\log2(9/14) \\ -(5/14)\log2(5/14) \\ =0.94$$

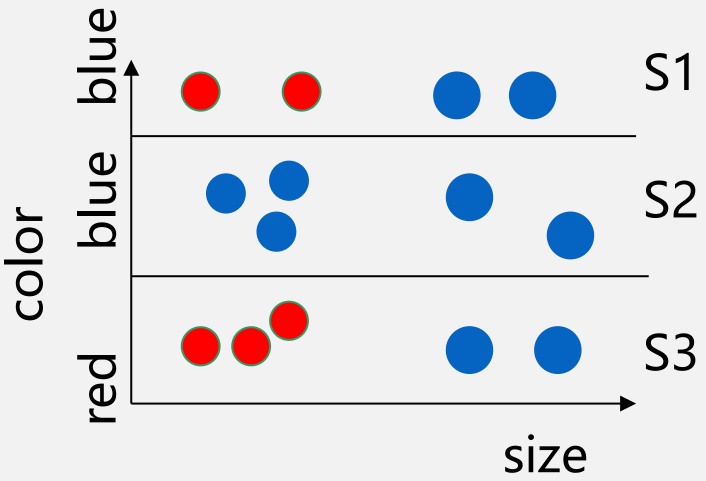
$$IG(S,X) = H(S) - \sum_v \left(\frac{S_v}{S}\right) H(S_v)$$

$$=0.94-4/14*1-5/14*0-5/14*0.971$$
$$=0.331$$

Information Gain: IG(X)



$IG(S,X)$ =0.395

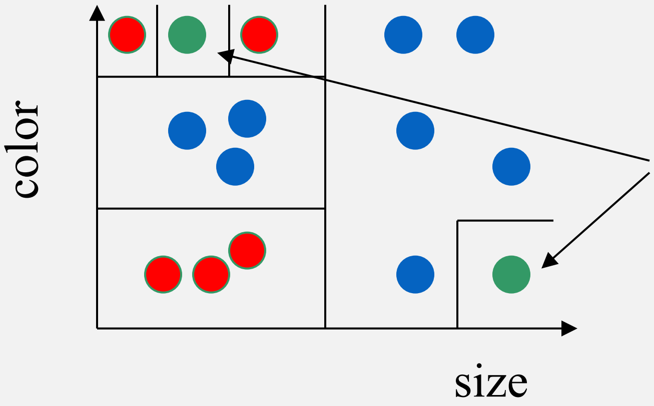


$IG(S,X)$ =0.331

∴ **The left partitioning schema is better.**

How deep should the tree be? Overfitting the Data

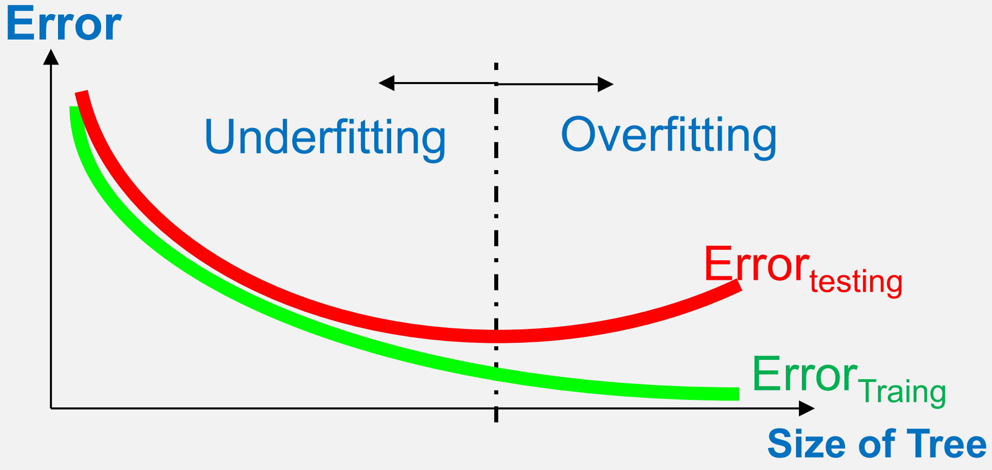
A tree **overfits** the data if we let it grow deep enough so that it begins to capture "aberrations" (偏差) in the data that harm the predictive power on unseen examples:



Possibly just noise, but the tree is grown deeper to capture these examples

Data Overfitting

A tree *t* in overfitting a dataset *D* if there is another tree ***t'*** where ***t*** has better classification accuracy than ***t'*** on *D* but worse classification accuracy than ***t'*** on D'.



Causes for Overfitting the Data

What causes a hypothesis to overfit the data?

- **Random errors or noise**

  Examples have incorrect class label or incorrect attribute values.

- **Coincidental patterns**

  By chance examples seem to deviate from a pattern due to the small size of the sample.

Overfitting is a serious problem that can cause serious performance degradation.

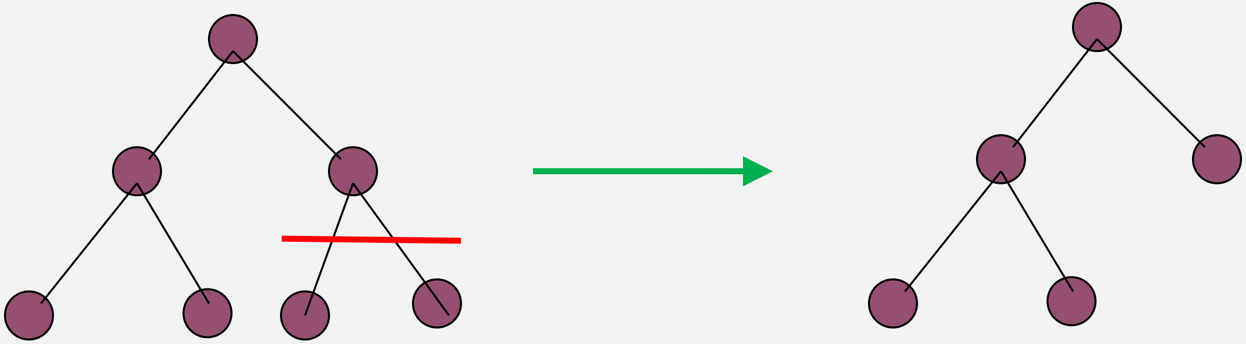
Solutions for Overfitting the Data

How to solve the overfitting data?

There are two main types of solutions:

- **Stop the tree early before it begins to overfit the data.**
  In practice, this solution is hard to implement because it is not clear what is a good stopping point.
- **Grow the tree until the algorithm stops even if the overfitting problem shows up. Then prune the tree as a post-processing step.**
  This method has found great popularity.

Decision Tree Pruning



Grow the tree to learn the training data

Prune tree to avoid overfitting the data

Typical Algorithms of Decision Tree

- **ID3：** Iterative Dichotomiser (二分法) 3, 1986, by *Ross Quinlan*
  - 其通过信息增益（Information Gain）选择最优特征进行节点分裂，递归构建决策树，只能用于离散特征处理，无缺失值处理，无剪枝
- **C4.5:** 1993， by *Ross Quinlan*
  - ID3的改进版, 引入增益率，支持连续特征和缺失值处理，加入剪枝
- **C5.0:** 1998， by *Ross Quinlan*
  - C4.5的改进版，改进了连续特征、缺失值和剪枝处理
- **CART:** Classification And Regression Tree, 1984， by *Leo Breiman*
  - 使用基尼指数（Gini Index）代替熵，支持回归任务

…

$$Gini(S) = 1 - \sum_{i=1}^{K} P_i^2$$

S：数据集
K：类别总数
p_i：第 i 类样本在数据集 s中的占比

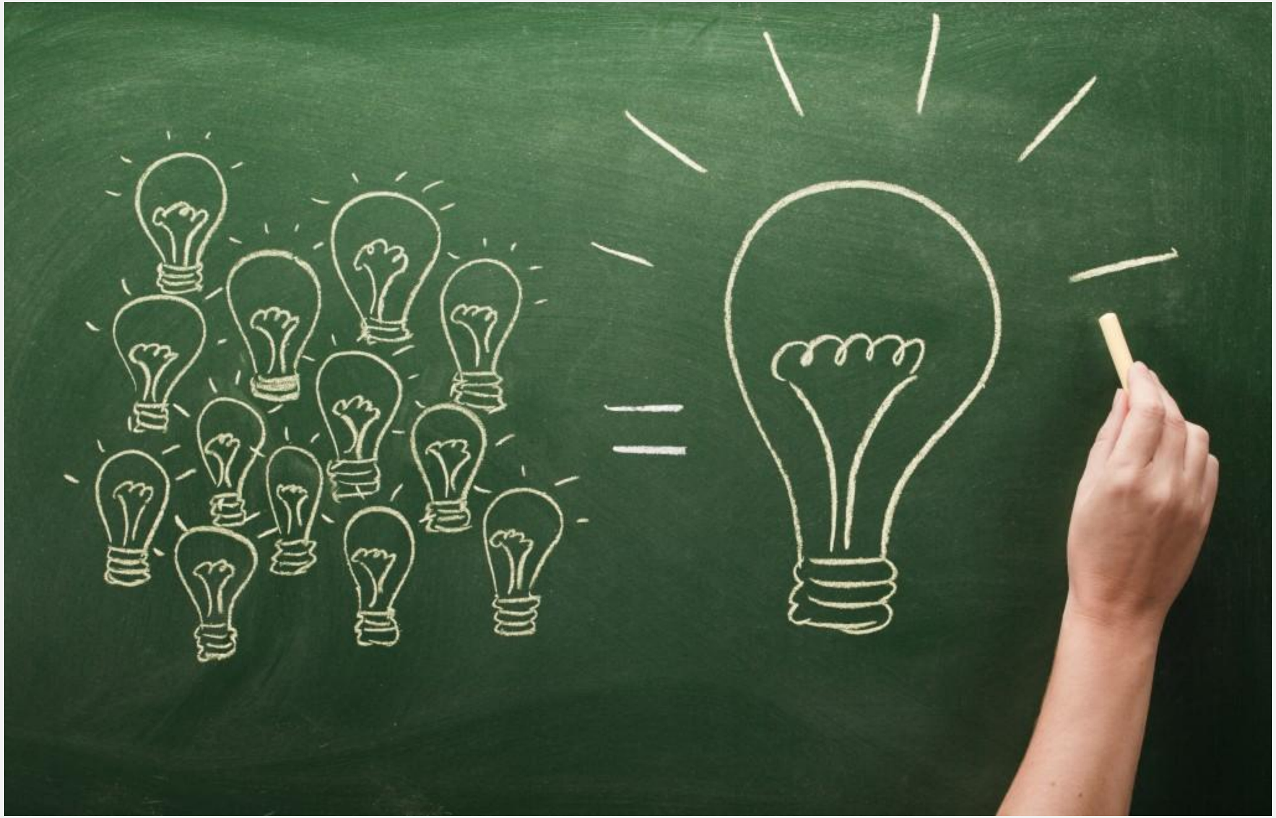
基尼指数用于反映数据集中类别的分布均匀程度：

- **值越小**，**数据纯度越高**（某个类别占主导）
- **值越大**，**数据越不纯**（类别分布越均匀）

Lect. 9 Machine Learning Classification

- What is Machine Learning
- Random Forest Classification
  - Decision Tree
  - Ensemble methods
  - Random Forest
- Cases

Power of the crowds



Ensemble methods

- A single decision tree is super fast
- But it does not perform well
- What if we learn multiple trees?

We need to make sure that not all trees just learn the same.

Ensemble Methods

- PAC 1984
- Weakly and Strongly Learning 1994

Theoretical Base

Boosting 1996

Bagging 1997

Random Forest 2001

Ensemble Learning

Probably Approximately Correct (PAC，概率近似正确理论) Learning

Two problems in ensemble learning:

How to train base classifiers?

How to fuse base classifiers?

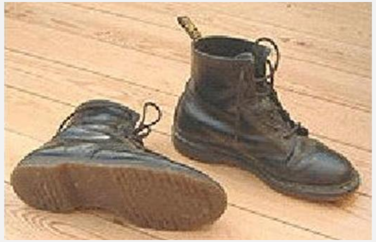
Boosting: Serial mode; Bagging: Parallel mode

Majority vote for classification； Average value for prediction

Ensemble Methods：Bagging

- If we split the data in randomly different ways, decision trees give different results, resulting in **high variance**.
  - **Bagging**: **B**ootstrap **agg**regat**ing** is a method that results in **low variance**.

Bootstrap:一种有放回的抽样方法



Pull up by your own bootstraps

- If we had multiple realizations of the data (or multiple samples) we could calculate the predictions multiple times and take the average of the fact that averaging multiple onerous estimations produce less uncertain results.
- Bagging通过从原始数据集中进行有放回的随机抽样来生成多个训练子集，每个子集都用于训练一个基分类器/基回归器,再通过集成这些基分类器/基回归器预测来提高整体预测的性能和稳定性。

The Process of Bagging

- Generate Bootstrap samples

  从数据集D中有放回地进行B次样本抽取，生成 B个Bootstrap 样本$\{D_1,D_2,\cdots,D_b,\cdots,D_B\}$，$B$通常取100-500.

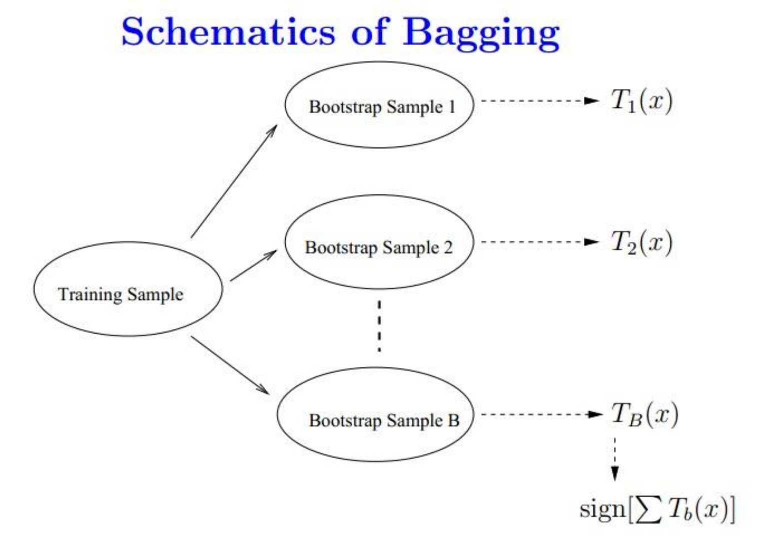
  未被抽中的样本为**OOB（Out-of-Bag）**

- Construct and train one base learner (decision tree, neural network, etc.) using each Bootstrap sample D_b

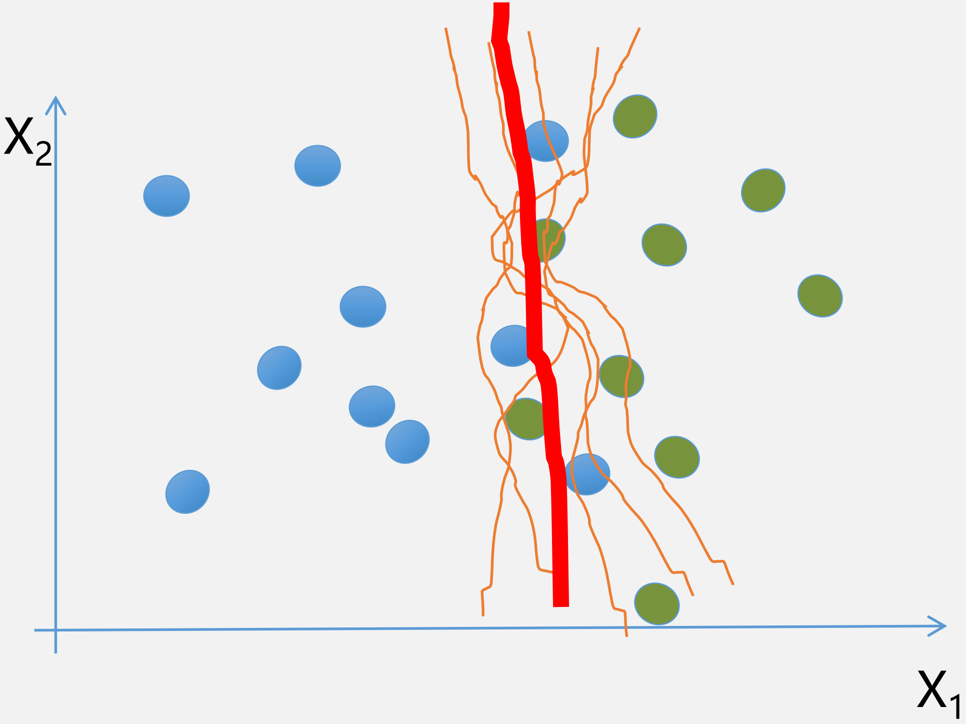
- Aggregate the results of all base learners
  - majority vote for classification
  - average for regression

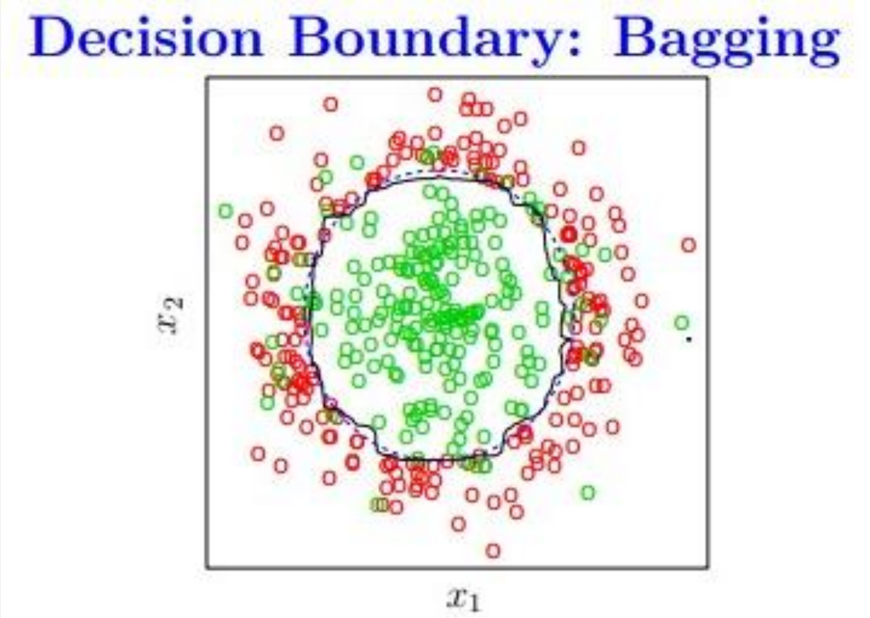
Bagging is Very effective

Random Forest is a typical type of Bagging ensamble method



Bagging for classification: Majority vote





***K*-means 无类处理这类数据**

Ensemble Methods：Boosting

Boosting is a general approach that can be applied to many statistical learning methods for regression or classification.

- 其每次使用的是全部的样本，每轮训练改变样本的权重
- 下一轮训练的目标是找到一个函数f 来拟合上一轮的残差
- 当残差足够小或者达到设置的最大迭代次数则停止

- **顺序训练**：基学习器按顺序生成，后续模型关注前序模型的错误。
- **权重调整**：根据前序模型的预测误差，动态调整样本权重或损失函数，使后续模型更关注难样本。
- **聚合方式**：通过**加权投票**或**加权求和**集成结果。

Boosting--Sequential fitting

Given the current model

- Fit a decision tree to the **residuals** from the model. Response variable now is the residuals and not Y.
- Then add this new decision tree into the fitted function in order to update the residuals.
- The learning rate has to be controlled .

XGBoost, daBoost, GBDT (Gradient Boosting Decision Tree), and LightGBM are Boosting algorithms

Lect. 9 Machine Learning
Classification

- What is Machine Learning
- **Random Forest Classification**
  - Decision Tree
  - Ensemble methods
  - **Random Forest**
- Cases

Radom Forest Proposed by Leo Breiman (1928-2005)

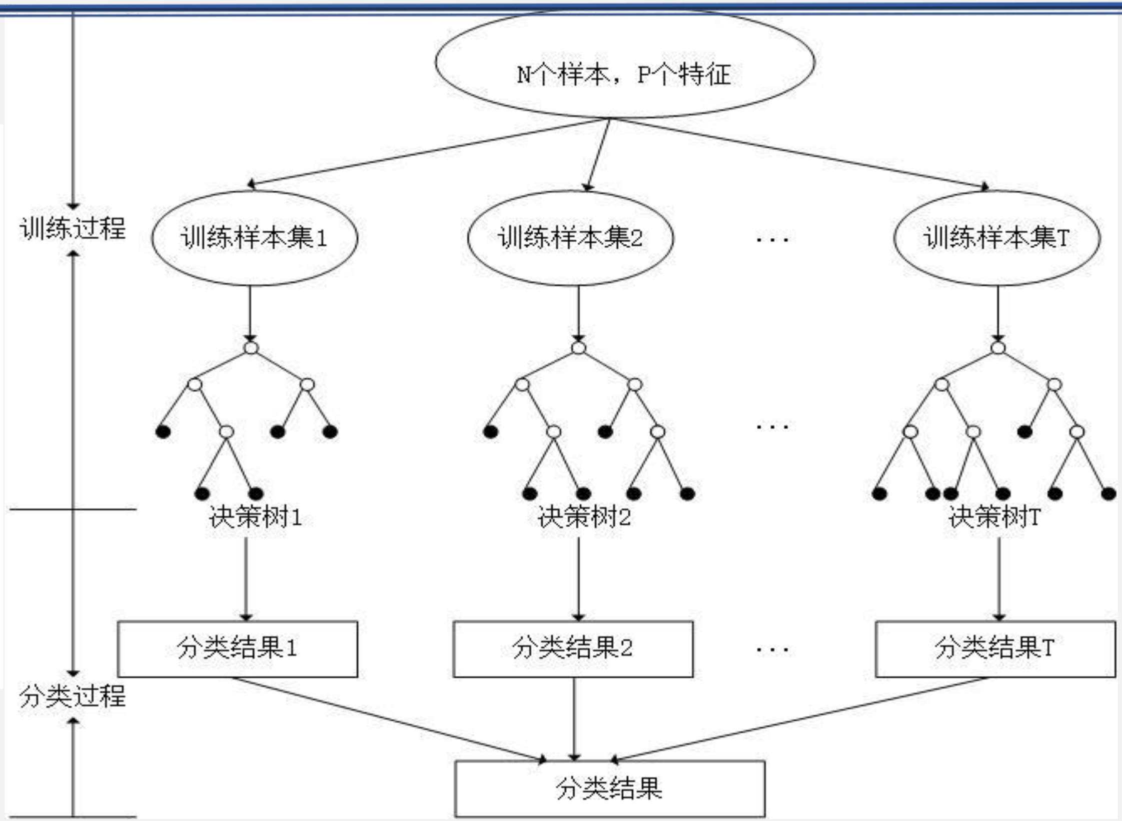
From Wikipedia

1954: PhD Berkeley (mathematics)
1960-1967: UCLA (mathematics)
1969-1982: Consultant
1982-1993: Berkeley (statistics)
1984: Classification And Regression Trees (CART)
1997: Bagging
2001: Radom Forest (Trees)

Leo Breiman. Random forests. Machine Learning, 2001, 45, 5–32.

Random forests are popular.
Leo Breiman's and Adele Cutler maintains a random forest website (http://www.stat.berkeley.edu/~breiman/RandomForests/) where the software is freely available, and of course it is included in every ML/STAT package.

Random Forest



Random Forest

- Random Forest (RF) is an ensemble learning technique based on a combination of sets of Classification and Regression Tree (CART) .
- Each tree is trained by sampling the training dataset and a set of variables independently through bagging aggregating .
- RF makes a tree grow by using the best split of a random subset of input features instead of using the best split variable, which can reduce the correlation between trees and the generalization error. Gini Index is usually used in RF as a measurement for the best split selection which maximizes dissimilarity between classes.
- RF presents many advantages of handling large dataset and variables but not of over-fitting.
- There are two parameters in the RF classifier: the number of active variables (M) in the random subset at each node and the number of trees (T) in the forest.

Refer to : Lu Lizhen, Tao Yuan and Di Liping. Object-Based Plastic-Mulched Landcover Extraction Using Integrated Sentinel-1 and Sentinel-2 Data. Remote Sens. 2018, 10(11), 1820; https://doi.org/10.3390/rs10111820.

Each Tree Grows as follows

- If the number of cases in the training set is *N*, sample *N* cases at random - but *with replacement*, from the original data. This sample will be the training set for growing the tree.
- If there are *M* input variables, a number $m<<M$ is specified such that at each node, *m* variables are selected at random out of the *M* and the best split on these *m* is used to split the node. The value of *m* is held constant during the forest growing.
- Each tree is grown to the largest extent possible. There is no pruning.

Random Forest Algorithm

- For Trees b = 1 to B:

  (a) Draw a bootstrap sample D_b of size N from the training data.

  (b) Grow a random-forest tree to the bootstrap sample D_b , by recursively repeating the following steps for each terminal node of the tree, until the minimum node size $\boldsymbol{n_{min}}$ is reached.

  - Select $\boldsymbol{m}$ variables at random from the M variables.
  - Pick the best variable/split-point among the m.
  - Split the node into two daughter nodes.

  Output the ensemble of trees.

- To make a prediction at a new point x we do:

  For regression: average the results

  For classification: majority vote

Random Forest Algorithm

In the original paper on random forests (*Leo Breiman. Random forests. Machine Learning, 2001, 45, 5–32*), it was shown that the forest error rate depends on two things:

- **The correlation between any two trees in the forest**. Increasing the correlation increases the forest error rate.
- **The strength of each individual tree in the forest**. A tree with a low error rate is a strong classifier. Increasing the strength of the individual trees decreases the forest error rate.

Reducing *m* *(the number of variables)* , reduces both the correlation and the strength. Increasing it increases both. Somewhere in between is an "optimal" range of *m* usually quite wide. Using the OOB (Out Of Bag) error rate, a value of *m* in the range can quickly be found. This is the only adjustable parameter to which random forests is somewhat sensitive.

Random Forest **Tuning**

- The inventors make the following recommendations:
  - For classification, the default value for ***m* is** $\sqrt{p}$ and **the minimum node size is one**.
  - For regression, the default value for ***m* is *p/3*** and **the minimum node size is five**.
- In practice, the best values for these parameters will depend on the problem, and they should be treated as tuning parameters.
- Like with Bagging, we can use OOB and therefore RF can be fit in one sequence, with cross-validation being performed along the way. Once the OOB error stabilizes, the training can be terminated.

Features and Advantages of Random Forest

The advantages of random forest are:

- It is one of the most accurate learning algorithms available. For many data sets, it produces a highly accurate classifier.
- It runs efficiently on large databases.
- It can handle thousands of input variables without variable deletion.
- It gives estimates of what variables are important in the classification.
- It generates an internal unbiased estimate of the generalization error as the forest building progresses.
- It has an effective method for estimating missing data and maintains accuracy when a large proportion of the data are missing.

Features and Advantages of Random Forest

- It has methods for balancing error in class population unbalanced data sets.
- Generated forests can be saved for future use on other data.
- Prototypes are computed that give information about the relation between the variables and the classification.
- It computes proximities between pairs of cases that can be used in clustering, locating outliers, or (by scaling) give interesting views of the data.
- The capabilities of the above can be extended to unlabeled data, leading to unsupervised clustering, data views and outlier detection.
- It offers an experimental method for detecting variable interactions.

Limitations of Random Forest

- Random forest have been observed to overfit for some datasets with noisy classification/regression tasks.
- For data including categorical variables with different number of levels, random forests are biased in favor of those attributes with more levels. Therefore, the variable importance scores from random forest are not reliable for this type of data.

Out-of-Bag Error Estimation

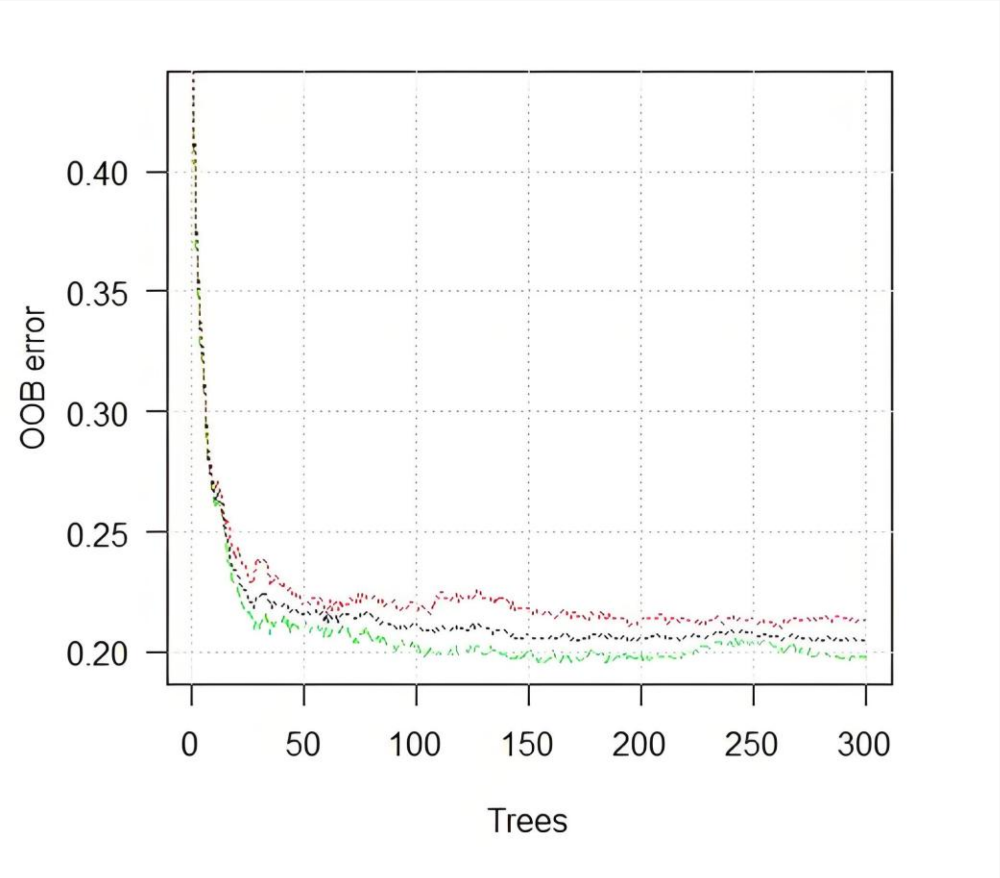
- Remember, in bootstrapping we sample with replacement, and therefore **not all observations are used for each bootstrap sample** (on average 1/3 of them are not used).
- We call them out-of-bag samples (OOB)
- We can predict the response for the *i-th* observation using each of the trees in which that observation was OOB and do this for *n* observations
- Calculate overall OOB MSE or classification error.

Out-of-Bag Error Estimation

In random forest, there is no need for cross-validation or a separate test set to get an unbiased estimate of the test set error. It is estimated internally, during the run, as follows:

- Each tree is constructed using a different bootstrap sample from the original data. About one-third of the cases are left out of the bootstrap sample and not used in the construction of the *k*th tree.
- Put each case left out in the construction of the *k*th tree down the *k*th tree to get a classification.
  - In this way, a test set classification is obtained for each case in about one-third of the trees.
  - At the end of the run, take *j* to be the class that got most of the votes every time case *n* was OOB.
  - The proportion of times that *j* is not equal to the true class of *n* averaged over all cases is the OOB error estimation.
  - This has proven to be unbiased in many tests.

Out-of-Bag Error Estimation Using Majority Vote



Variable importance

- In every tree grown in the forest, put down the OOB cases and count the number of votes cast for the correct class. Now randomly permute the values of variable *m* in the OOB cases and put these cases down the tree. Subtract the number of votes for the correct class in the variable-*m*-permuted OOB data from the number of votes for the correct class in the untouched OOB data. The average of this number over all trees in the forest is the raw importance score for variable m.
- If the values of this score from tree to tree are independent, then the standard error can be computed by a standard computation. The correlations of these scores between trees have been computed for a number of data sets and proved to be quite low, therefore we compute standard errors in the classical way, divide the raw score by its standard error to get a z-score, ands assign a significance level to the z-score assuming normality.
- If the number of variables is very large, random forest can be run once with all the variables, then run again using only the most important variables from the first run.
- For each case, consider all the trees for which it is OOB. Subtract the percentage of votes for the correct class in the variable-m-permuted OOB data from the percentage of votes for the correct class in the untouched OOB data. This is the local importance score for variable *m* for this case, and is used in the graphics program RAFT.

Variable importance

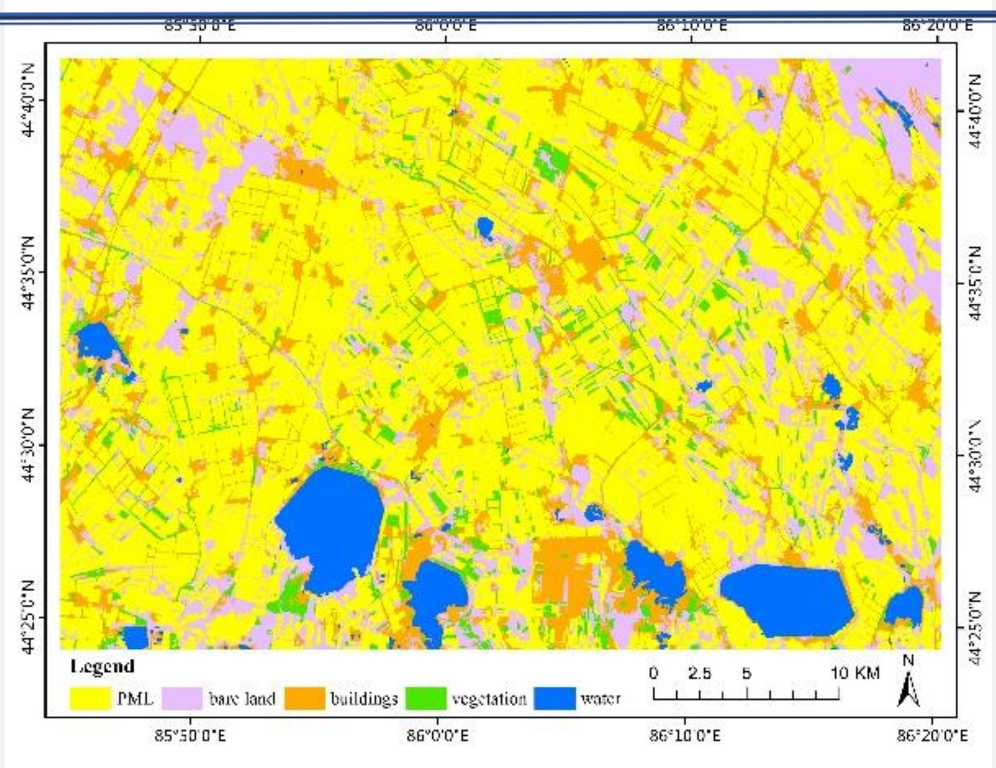
Lu et al. Object-Based Plastic-Mulched Landcover Extraction Using Integrated Sentinel-1 and Sentinel-2 Data. Remote Sens. 2018, 10(11), 1820.

S1 and S2		S1		S2	
Features	Importance	Features	Importance	Features	Importance
Mean band 1	0.13	Mean VH_db	0.22	NDVI	0.11
Mean band 8	0.10	Mean VH/VV	0.14	Mean band 8	0.11
NDVI	0.10	Mean VV_db	0.11	GLCM-d	0.10
SD band 2	0.09	SD VV_db	0.09	Mean band 1	0.09
SD VV_db	0.09	GLCM-e-VV_db	0.08	SD band 3	0.09
GLCM-d	0.08	GLCM-h-VV_db	0.07	Mean band 3	0.08
Mean band 3	0.07	GLCM-c-VH_db	0.07	GLCM-c-8	0.06
Mean VH/VV	0.07	GLCM-d-VH_db	0.05	SD band 12	0.05
SD band 5	0.06	GLCM-e-VH/VV	0.05	PMLI	0.05
Mean band 4	0.04	SD VH_db	0.04	GLCM-h	0.05

Iterations

- The operating definition of interaction used is that variables *m* and *k* interact if a split on one variable, say *m*, in a tree makes a split on *k* either systematically less possible or more possible. The implementation used is based on the GINI values *g(m)* for each tree in the forest. These are ranked for each tree and for each two variables, the absolute difference of their ranks are averaged over all trees.
- This number is also computed under the hypothesis that the two variables are independent of each other and the latter subtracted from the former. A large positive number implies that a split on one variable inhibits a split on the other and conversely. This is an experimental procedure whose conclusions need to be regarded with caution. It has been tested on only a few data sets.

The Classification Results of RF



Object-based Sentinel-1 classification results with random forest classifier

Evaluation Metrics

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}$$

$$\text{kappa} = \frac{2\cdot(\text{TP}\cdot\text{TN} - \text{FN}\cdot\text{FP})}{(\text{TP} + \text{FP})\cdot(\text{FP} + \text{TN}) + (\text{TP} + \text{FN})\cdot(\text{FN} + \text{TN})}$$

Precision is the fraction of correct positives (TP) among the total predicted positives (TP + FP), and recall is the fraction of correct positives (TP) among the total existing positives (TP + FN). Both metrics make it possible to assess model performance on the minority class (i.e., BUA) (He and Ma, 2013). As a consequence, the joint analysis of pre-

Lect. 9 Machine Learning Classification

- ☐ What is Machine Learning
- ☐ Random Forest Classification
  - ➢ Decision Tree
  - ➢ Ensemble methods
  - ➢ Random Forest
- ☐ **Cases**

Cases

- Mapping global water-surface photovoltaics with satellite images-2023
- Mapping crop type in Northeast China during 2013-2021 using automatic sampling and tile-based image classification-2023
- Seamless and automated rapeseed mapping for large cloudy regions using time-series optical satellite imagery-2024
- Characterizing land use changes triggered by crop-aquaculture co-cultivation from 2013 to 2022 based on a robust classification framework: Illustration in Jianghan Plain, China，2026，RSE

References

1. Leo Breiman. Random forests. Mach. Learn. 2001, 45, 5–32.
2. https://www.stat.berkeley.edu/~breiman/RandomForests/cc_home.htm#overview
3. ArcGIS Help.
4. Lu Lizhen, et al. A Decision-tree Classifier for Extracting Transparent Plastic-mulched Landcover from Landsat-5 TM Images. IEEE Journal of Selected Topics in Applied Earth Observation and Remote Sensing. 2014, 7(11):4548-4558.
5. Lu Lizhen, et al. Object-Based Plastic-Mulched Landcover Extraction Using Integrated Sentinel-1 and Sentinel-2 Data. Remote Sens. 2018, 10(11), 1820; https://doi.org/10.3390/rs10111820.
6. Mapping global water-surface photovoltaics with satellite images-2023
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9. Characterizing land use changes triggered by crop-aquaculture co-cultivation from 2013 to 2022 based on a robust classification framework: Illustration in Jianghan Plain, China，2026，RSE
10. A Google Earth Engine Approach for Wildfire Susceptibility Prediction Fusion with Remote Sensing Data of Different Spatial Resolutions-2022

Reading , Exercises & Lab

1. Read at least on paper on RF.
2. What is decision tree? List more than 3 typical algorithms of decision tree.
3. What is Random Forest (RF)? List more than 3 application cases of RF, and try to describe some disadvantages and advantages of RF.
4. Do Lab 9.

Lab report should be submitted before 20:00,May. 8th